CSE 464 – Software QA and Testing
Project Part 2 Report

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Course: CSE 464
Date: April 3, 2026

GitHub Repository:
https://github.com/vamika27/CSE4642026-vnegi3

Project Overview

This report documents the implementation of Project Part 2, which extends the directed graph processing system from Part 1 with the following new features:

- Remove node, remove nodes, and remove edge APIs with test cases
- Continuous integration using GitHub Actions
- BFS graph search implemented on a dedicated branch
- DFS graph search implemented on a dedicated branch
- Merging both branches into main with conflict resolution using an Algorithm enum

How to Build and Run

Clone the repository:

```
git clone https://github.com/vamika27/CSE4642026-vnegi3
```

Navigate to the project directory:

```
cd CSE4642026-vnegi3
```

Compile and run all tests:

```
mvn package
```

Run tests only:

```
mvn test
```

Sample input file:

sample_inputs/input1.dot

Project Structure

```
CSE4642026-vnegi3
├── pom.xml
├── README.md
├── .github
│   └── workflows
│       └── ci.yml
├── sample_inputs
│   └── input1.dot
└── src
    ├── main
    │   └── java
    │       └── graph
    │           ├── Algorithm.java
    │           ├── Graph.java
    │           ├── GraphParser.java
    │           └── Path.java
    └── test
        └── java
            └── graph
                └── GraphTest.java
```

Feature 5 – Remove Node, Remove Nodes, and Remove Edge APIs

Three new APIs were added to the Graph class to support removing nodes and edges.

API: removeNode(String label)

This method removes a single node from the graph. It also removes all outgoing edges from that node and all incoming edges to that node from other nodes. If the specified node does not exist in the graph, the method throws a NoSuchElementException.

API: removeNodes(String[] labels)

This method removes multiple nodes from the graph by iterating through the provided array and calling removeNode() for each label. If any node in the array does not exist, a

NoSuchElementException is thrown.

API: removeEdge(String srcLabel, String dstLabel)

This method removes a single directed edge from the source node to the destination node. The nodes themselves remain in the graph. If either node does not exist, or if the specified edge does not exist, a NoSuchElementException is thrown.

Test cases cover three scenarios as required by the assignment:

Scenario 1: Nodes and edges are correctly removed.
- testRemoveNode: Parses input1.dot, removes node "a", verifies node count is 7, verifies outgoing edges from "a" and incoming edge d->a are removed.
- testRemoveNodes: Removes nodes "f" and "g", verifies node count is 6 and edges e->f and e->g are removed.
- testRemoveEdge: Removes edge a->b, verifies edge count decreases by 1 while both nodes "a" and "b" still exist.

Scenario 2: Removing non-existent nodes throws exceptions.
- testRemoveNonExistentNode: Attempts to remove node "z" from a graph that only contains node "a". Verifies NoSuchElementException is thrown.
- testRemoveNodesOneNonExistent: Attempts to remove ["a", "nonexistent"]. Verifies NoSuchElementException is thrown.

Scenario 3: Removing non-existent edges throws exceptions.
- testRemoveNonExistentEdge: Attempts to remove edge a->c when only a->b exists. Verifies NoSuchElementException is thrown.
- testRemoveEdgeWithNonExistentNode: Attempts to remove edge a->z when node "z" does not exist. Verifies NoSuchElementException is thrown.

GitHub commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/ea7cc9194b137f14ce91b1108da73404faa20cc6

```
Last login: Fri Apr  3 17:30:46 on ttys001
[vamikanegi@Vamikas-MacBook-Air ~ % cd CSE4642026-vnegi3
cd: no such file or directory: CSE4642026-vnegi3
[vamikanegi@Vamikas-MacBook-Air ~ % cd CSE464
[vamikanegi@Vamikas-MacBook-Air CSE464 % cd CSE4642026-vnegi3
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git checkout main
M       README.md
Switched to branch 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git pull origin main

remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 5.51 KiB | 940.00 KiB/s, done.
From https://github.com/vamika27/CSE4642026-vnegi3
 * branch            main       -> FETCH_HEAD
   9d0e7eb..c77b2df  main       -> origin/main
Updating da0f05b..c77b2df
error: Your local changes to the following files would be overwritten by merge:
        README.md
Please commit your changes or stash them before you merge.
Aborting
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< asu.cse464:CSE4642026-vnegi3 >--------------------
[INFO] Building CSE 464 Course Project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-vnegi3 ---
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 2 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-vnegi3 ---
[INFO] skip non existing resourceDirectory /Users/vamikanegi/CSE464/CSE4642026-vnegi3/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-vnegi3 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running graph.GraphTest
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.094 s -- in graph.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-vnegi3 ---
[INFO] Building jar: /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.906 s
[INFO] Finished at: 2026-04-03T17:44:08-07:00
[INFO] ------------------------------------------------------------------------
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git add .
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git commit -m "Feature 5: add removeNode, removeNodes, removeEdge APIs and tests"
[main ea7cc91] Feature 5: add removeNode, removeNodes, removeEdge APIs and tests
 14 files changed, 527 insertions(+), 18 deletions(-)
 create mode 100644 .DS_Store
 create mode 100644 CSE 464 course project 1 Vamika Negi.pdf
 create mode 100644 sample_inputs/graph_output.dot
 create mode 100644 sample_inputs/graph_output.png
 create mode 100644 sample_inputs/output1.dot
 create mode 100644 sample_inputs/outputGraph.txt
 create mode 100644 src/.DS_Store
 create mode 100644 src/main/.DS_Store
 create mode 100644 src/main/java/.DS_Store
 create mode 100644 src/test/.DS_Store
 create mode 100644 src/test/java/.DS_Store
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git push origin main
```

Feature 6 – Continuous Integration with GitHub Actions

A GitHub Actions workflow was added to automatically build and test the project whenever code is pushed to the main, bfs, or dfs branches, or when a pull request is opened against main.

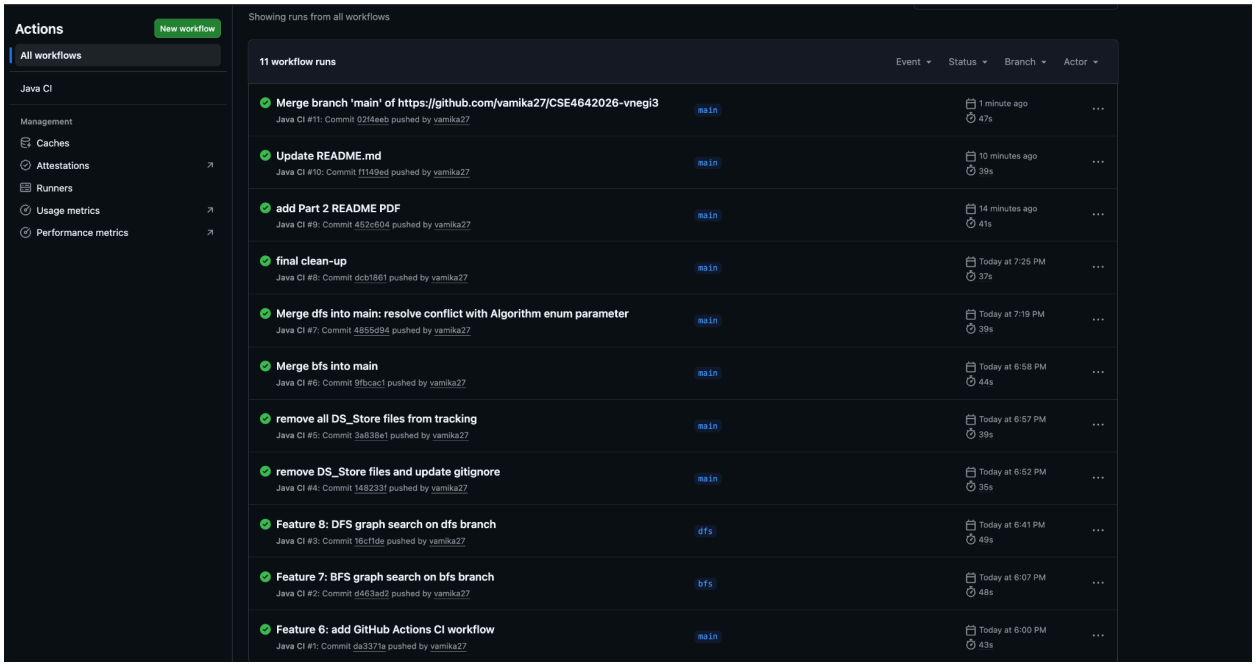
The workflow file is located at .github/workflows/ci.yml and performs the following steps:

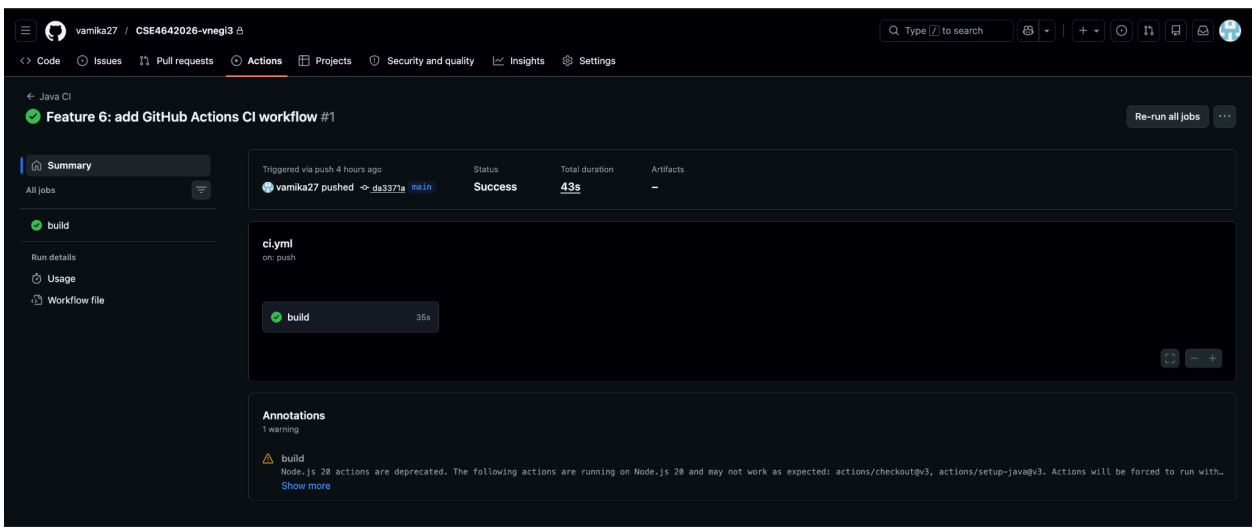
1. Checks out the repository code
2. Sets up JDK 17 using the temurin distribution
3. Installs Graphviz (required for the testOutputGraphics test)
4. Builds and tests the project using mvn package

The workflow runs on ubuntu-latest and triggers on every push to ensure that all code changes are automatically validated.

GitHub commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/da3371afe9b62cc8f35f1354d10c7f1ee099ea0a

Feature 6 in actions tab:





Feature 6 workflow run

Feature 7 – BFS Graph Search (bfs branch)

A new branch called "bfs" was created from main to implement a graph search API using the Breadth-First Search algorithm.

API: Path GraphSearch(String src, String dst)

This method finds a path from a source node to a destination node using BFS. It uses a Queue (LinkedList) to explore nodes level by level. A parent map tracks the path from source to

destination. If a path exists, the method reconstructs it and returns a Path object. If no path exists, or if either node is not in the graph, the method returns null.

A new Path class was created to represent a path through the graph. The Path class stores an ordered list of node labels and provides a toString() method that formats the path as n1 -> n2 -> n3.

Test cases for BFS:
- testBFSPathExists: Searches from "a" to "h" in input1.dot. Verifies a path is found that starts at "a" and ends at "h".
- testBFSPathNotExists: Searches from "h" to "a". Since "h" has no outgoing edges, verifies null is returned.
- testBFSSameNode: Searches from "a" to "a". Verifies a path of length 1 is returned.
- testBFSNodeNotInGraph: Searches from "a" to "z" where "z" does not exist. Verifies null is returned.

Branch: https://github.com/vamika27/CSE4642026-vnegi3/tree/bfs

GitHub commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/d463ad2ff6f3c6820113b2dcc77a439f894f2322

```
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mkdir -p .github/workflows
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % nano .github/workflows/ci.yml
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git add .
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git commit -m "Feature 6: add GitHub Actions CI workflow"
 [main da3371a] Feature 6: add GitHub Actions CI workflow
  5 files changed, 26 insertions(+)
  create mode 100644 .github/workflows/ci.yml
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git push origin main
 Enumerating objects: 20, done.
 Counting objects: 100% (20/20), done.
 Delta compression using up to 8 threads
 Compressing objects: 100% (10/10), done.
 Writing objects: 100% (12/12), 1.50 KiB | 1.50 MiB/s, done.
 Total 12 (delta 5), reused 0 (delta 0), pack-reused 0
 remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
 To https://github.com/vamika27/CSE4642026-vnegi3.git
    ea7cc91..da3371a  main -> main
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git checkout main
 Already on 'main'
 Your branch is up to date with 'origin/main'.
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git checkout -b bfs
 Switched to a new branch 'bfs'
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mvn package
 [INFO] Scanning for projects...
 [INFO]
 [INFO] -------------------< asu.cse464:CSE4642026-vnegi3 >--------------------
 [INFO] Building CSE 464 Course Project 1.0-SNAPSHOT
 [INFO]   from pom.xml
 [INFO] --------------------------------[ jar ]---------------------------------
 [INFO]
 [INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-vnegi3 ---
 [INFO] Copying 0 resource from src/main/resources to target/classes
 [INFO]
 [INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-vnegi3 ---
 [INFO] Recompiling the module because of changed source code.
 [INFO] Compiling 3 source files with javac [debug target 17] to target/classes
 [WARNING] location of system modules is not set in conjunction with -source 17
   not setting the location of system modules may lead to class files that cannot run on JDK 17
     --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
 [INFO]
 [INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-vnegi3 ---
 [INFO] skip non existing resourceDirectory /Users/vamikanegi/CSE464/CSE4642026-vnegi3/src/test/resources
 [INFO]
 [INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-vnegi3 ---
 [INFO] Recompiling the module because of changed dependency.
 [INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
 [WARNING] location of system modules is not set in conjunction with -source 17
   not setting the location of system modules may lead to class files that cannot run on JDK 17
     --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
 [INFO]
 [INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-vnegi3 ---
 [INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
 [INFO]
 [INFO] -------------------------------------------------------
 [INFO]  T E S T S
 [INFO] -------------------------------------------------------
 [INFO] Running graph.GraphTest
 [INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.226 s -- in graph.GraphTest
 [INFO]
 [INFO] Results:
 [INFO]
 [INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
 [INFO]
 [INFO]
 [INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-vnegi3 ---
 [INFO] Building jar: /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar
 [INFO] ------------------------------------------------------------------------
 [INFO] BUILD SUCCESS
 [INFO] ------------------------------------------------------------------------
 [INFO] Total time:  2.032 s
 [INFO] Finished at: 2026-04-03T18:06:40-07:00
 [INFO] ------------------------------------------------------------------------
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git add .
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git commit -m "Feature 7: BFS graph search on bfs branch"
 [bfs d463ad2] Feature 7: BFS graph search on bfs branch
  9 files changed, 113 insertions(+), 13 deletions(-)
  create mode 100644 src/main/java/graph/Path.java
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git push origin bfs
 Enumerating objects: 33, done.
 Counting objects: 100% (33/33), done.
 Delta compression using up to 8 threads
 Compressing objects: 100% (17/17), done.
 Writing objects: 100% (18/18), 2.47 KiB | 2.47 MiB/s, done.
 Total 18 (delta 9), reused 0 (delta 0), pack-reused 0
 remote: Resolving deltas: 100% (9/9), completed with 8 local objects.
 remote:
 remote: Create a pull request for 'bfs' on GitHub by visiting:
 remote:      https://github.com/vamika27/CSE4642026-vnegi3/pull/new/bfs
```

All 18 tests pass in feature 7

Feature 8 – DFS Graph Search (dfs branch)

A new branch called "dfs" was created from main to implement the same graph search API using the Depth-First Search algorithm.

API: Path GraphSearch(String src, String dst)

This method finds a path from a source node to a destination node using DFS. It uses a Stack instead of a Queue to explore nodes depth-first. The parent map and path reconstruction logic are identical to the BFS implementation. If a path exists, the method returns a Path object. If no path exists, or if either node is not in the graph, the method returns null.

Test cases for DFS:
- testDFSPathExists: Searches from "a" to "h" in input1.dot. Verifies a path is found that starts at "a" and ends at "h".
- testDFSPathNotExists: Searches from "h" to "a". Verifies null is returned.
- testDFSSameNode: Searches from "a" to "a". Verifies a path of length 1 is returned.
- testDFSNodeNotInGraph: Searches from "a" to "z" where "z" does not exist. Verifies null is returned.

Branch: https://github.com/vamika27/CSE4642026-vnegi3/tree/dfs

GitHub commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/16cf1dee54f586ba6998d26cc5395bd561b6a039

```
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< asu.cse464:CSE4642026-vnegi3 >-------------------
[INFO] Building CSE 464 Course Project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-vnegi3 ---
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-vnegi3 ---
[INFO] skip non existing resourceDirectory /Users/vamikanegi/CSE464/CSE4642026-vnegi3/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-vnegi3 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running graph.GraphTest
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.254 s -- in graph.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-vnegi3 ---
[INFO] Building jar: /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.113 s
[INFO] Finished at: 2026-04-03T18:39:39-07:00
[INFO] ------------------------------------------------------------------------
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git add .
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git commit -m "Feature 8: DFS graph search on dfs branch"
[dfs 16cf1de] Feature 8: DFS graph search on dfs branch
 9 files changed, 110 insertions(+), 13 deletions(-)
 create mode 100644 src/main/java/graph/Path.java
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git push origin dfs
```

Feature 9 – Merge BFS and DFS Branches with Conflict Resolution

The bfs branch was first merged into main cleanly. Then the dfs branch was merged into main, which produced a merge conflict in Graph.java and GraphTest.java because both branches added a GraphSearch method with the same signature but different implementations.

The conflict was resolved by introducing a new Algorithm enum and changing the method signature to accept a third parameter for choosing the search algorithm.

New file: Algorithm.java

```
public enum Algorithm {
    BFS,
    DFS
```

```
}
```

Updated API: Path GraphSearch(String src, String dst, Algorithm algo)

This unified method finds a path from src to dst using the specified algorithm. Internally, it uses a Deque as the data structure. When Algorithm.BFS is specified, nodes are removed from the front of the deque (queue behavior). When Algorithm.DFS is specified, nodes are removed from the back of the deque (stack behavior).

Example usage:

```
Graph graph = GraphParser.parseGraph("sample_inputs/input1.dot");
Path bfsPath = graph.GraphSearch("a", "h", Algorithm.BFS);
Path dfsPath = graph.GraphSearch("a", "h", Algorithm.DFS);
System.out.println("BFS: " + bfsPath);  // BFS: a -> e -> f -> h
System.out.println("DFS: " + dfsPath);  // DFS: a -> e -> g -> h
```

The final test suite contains 22 tests covering all features from Part 1 and Part 2, including both BFS and DFS search tests using the Algorithm enum.

Merge bfs into main commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/9fbcac15db00cf76d8efd60d959b7277a7690d63

Merge dfs into main (conflict resolution) commit:
https://github.com/vamika27/CSE4642026-vnegi3/commit/4855d94ed102097274f0e3b9455086ffdd0fd702

```
[vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< asu.cse464:CSE4642026-vnegi3 >--------------------
[INFO] Building CSE 464 Course Project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-vnegi3 ---
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 4 source files with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-vnegi3 ---
[INFO] skip non existing resourceDirectory /Users/vamikanegi/CSE464/CSE4642026-vnegi3/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-vnegi3 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
  not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-vnegi3 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running graph.GraphTest
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.222 s -- in graph.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-vnegi3 ---
[INFO] Building jar: /Users/vamikanegi/CSE464/CSE4642026-vnegi3/target/CSE4642026-vnegi3-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.983 s
[INFO] Finished at: 2026-04-03T19:19:25-07:00
[INFO] ------------------------------------------------------------------------
vamikanegi@Vamikas-MacBook-Air CSE4642026-vnegi3 % git add .
git commit -m "Merge dfs into main: resolve conflict with Algorithm enum parameter"
[git push origin main
[main 4855d94] Merge dfs into main: resolve conflict with Algorithm enum parameter
Enumerating objects: 45, done.
Counting objects: 100% (45/45), done.
Delta compression using up to 8 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (18/18), 1.95 KiB | 1.95 MiB/s, done.
Total 18 (delta 9), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (9/9), completed with 8 local objects.
To https://github.com/vamika27/CSE4642026-vnegi3.git
   9fbcac1..4855d94  main -> main
```

All 22 tests pass

Unit Testing Summary

The final test suite contains 22 tests:

From Part 1 (7 tests):
- testParseGraph
- testAddNode
- testAddNodes
- testAddEdge
- testOutputGraph

- testOutputDOTGraph
- testOutputGraphics

Remove API tests (7 tests):
- testRemoveNode
- testRemoveNodes
- testRemoveEdge
- testRemoveNonExistentNode
- testRemoveNodesOneNonExistent
- testRemoveNonExistentEdge
- testRemoveEdgeWithNonExistentNode

BFS search tests (4 tests):
- testBFSPathExists
- testBFSPathNotExists
- testBFSSameNode
- testBFSNodeNotInGraph

DFS search tests (4 tests):
- testDFSPathExists
- testDFSPathNotExists
- testDFSSameNode
- testDFSNodeNotInGraph

All 22 tests pass with 0 failures and 0 errors.

Commit Summary

Feature 5 – Remove APIs and tests:
https://github.com/vamika27/CSE4642026-vnegi3/commit/ea7cc9194b137f14ce91b1108da73404faa20cc6

Feature 6 – GitHub Actions CI:
https://github.com/vamika27/CSE4642026-vnegi3/commit/da3371afe9b62cc8f35f1354d10c7f1ee099ea0a

Feature 7 – BFS graph search (bfs branch):
https://github.com/vamika27/CSE4642026-vnegi3/commit/d463ad2ff6f3c6820113b2dcc77a439f894f2322

Feature 8 – DFS graph search (dfs branch):
https://github.com/vamika27/CSE4642026-vnegi3/commit/16cf1dee54f586ba6998d26cc5395bd561b6a039

Merge bfs into main:
https://github.com/vamika27/CSE4642026-vnegi3/commit/9fbcac15db00cf76d8efd60d959b7277a7690d63

Merge dfs into main (conflict resolution with Algorithm enum):
https://github.com/vamika27/CSE4642026-vnegi3/commit/4855d94ed102097274f0e3b9455086ffdd0fd702

Branch links:
bfs: https://github.com/vamika27/CSE4642026-vnegi3/tree/bfs
dfs: https://github.com/vamika27/CSE4642026-vnegi3/tree/dfs

GitHub Actions:
https://github.com/vamika27/CSE4642026-vnegi3/actions